Texture-aware Intrinsic Image Decomposition with Model- and Learning-based Priors

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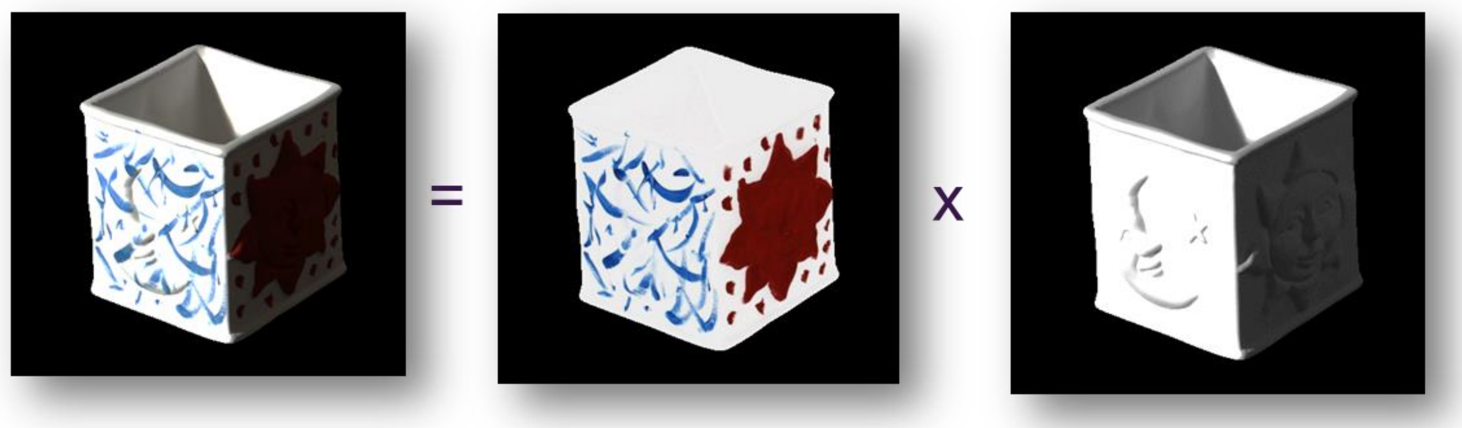
Highlights

Introduction

- Intrinsic image decomposition aims to separate an image **I** into reflectance layer **R** and shading layer **S** as **I = R ⊙ S**.
- Shading image contains information about illumination conditions and scene geometry, reflectance image captures the material's reflectance properties.

Challenges

- Model-based methods fail to disentangle the spatially-varying lighting and rich textures.
- Learning-based method is able to separate the lighting effect from reflectance layer but ends up with missing textures in intrinsic images.



Input image　Reflectance　Shading

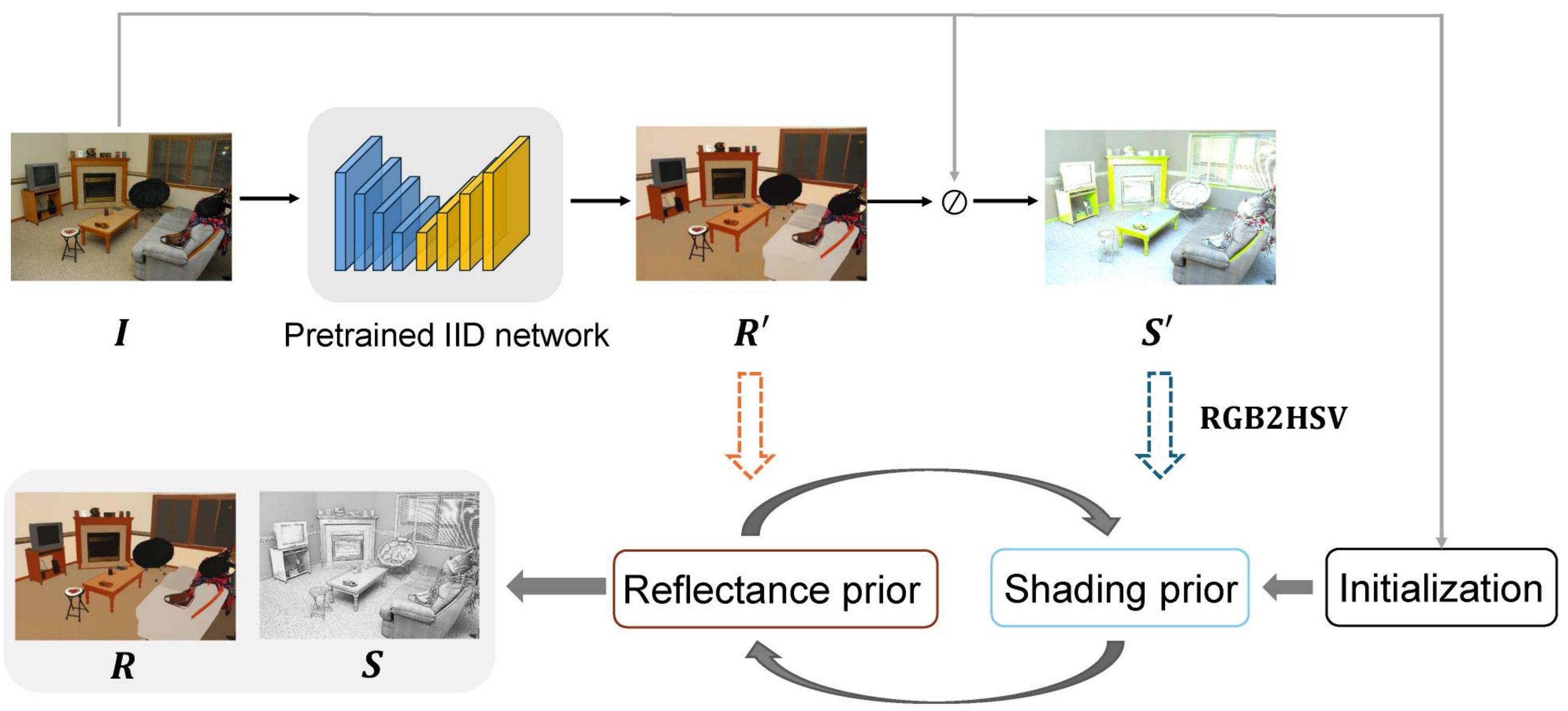
Motivation of Our Solutions

We propose to inherit the merits of the model-based method and learning-based method by integrating the learning-based prior into model-based framework.

Our Solution

Our Pipeline

- We design a learning-based prior to regularize the reflectance and shading term in traditional optimization framework.
- A smooth reflectance prior and a texture-aware shading prior (extracted from pretrained network) are integrated into optimization framework.



Model Solver

- We formulate the ensemble energy function as:

$$E(\mathbf{S},\mathbf{R}) = \|\mathbf{R}\odot\mathbf{S}-\mathbf{I}\|_2^2 + \alpha\|\mathbf{TV}(\mathbf{S}-\mathbf{S}')\|_1 + \beta\|\nabla\mathbf{R}\|_0 + \gamma\|\mathbf{S}-\mathbf{S}_0\|_2^2, \quad \text{s.t.} \quad \mathbf{I}\leq\mathbf{S}$$

- We then apply alternating projection to split the optimization into two subproblems with respect to R and S.

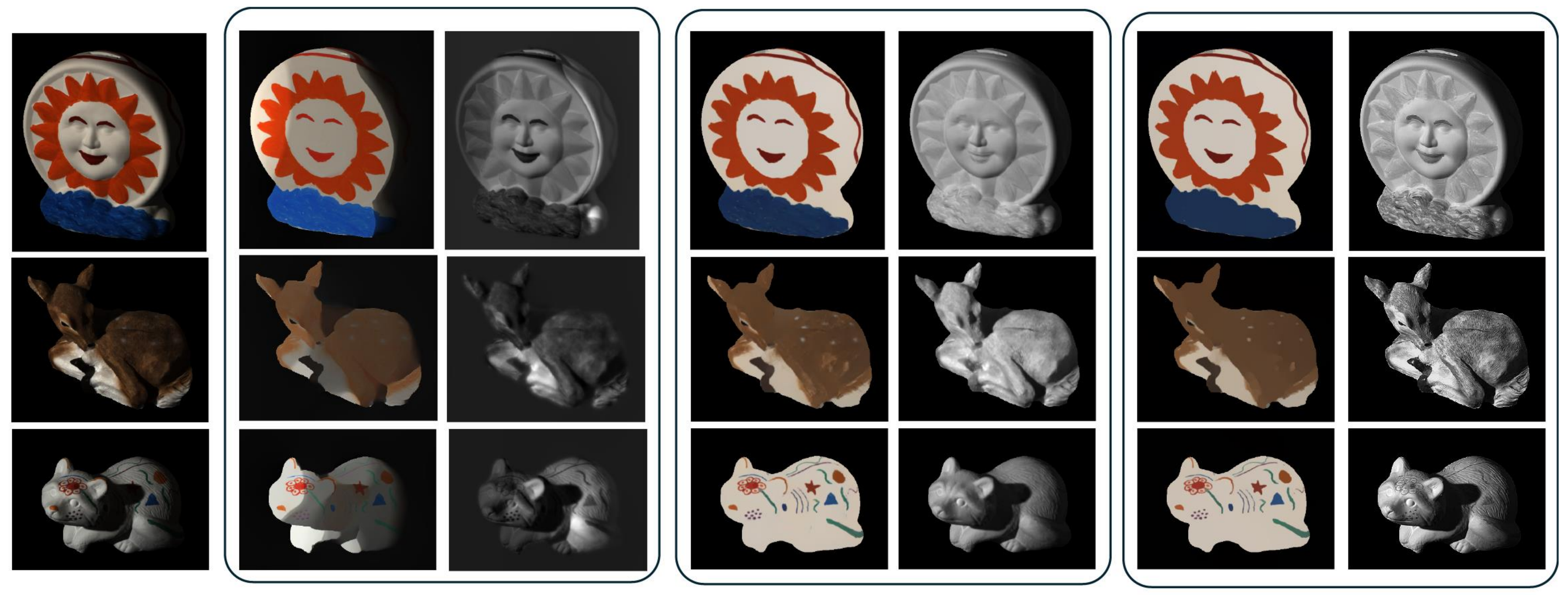
$$\underset{\mathbf{R}}{\operatorname{argmin}} \ \|\mathbf{R}\odot\mathbf{S}-\mathbf{I}\|_2^2 + \beta\|\nabla\mathbf{R}\|_0,$$

$$\underset{\mathbf{S}}{\operatorname{argmin}} \ \|\mathbf{R}\odot\mathbf{S}-\mathbf{I}\|_2^2 + \alpha\|\mathbf{TV}(\mathbf{S}-\mathbf{S}')\|_1 + \gamma\|\mathbf{S}-\mathbf{S}_0\|_2^2.$$

Which can be solved via closed-form solution.

Experimental Results

Visualization Results



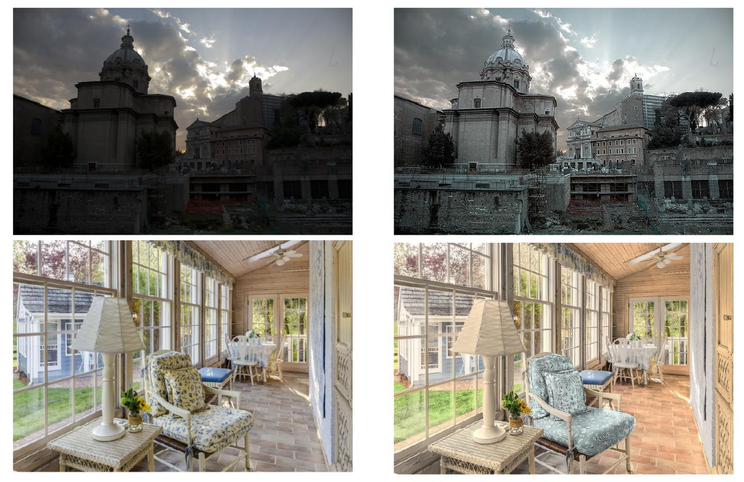
(a) Input　(b) Fu et al. [11]　(c) CRefNet [29]　(d) Ours

Texture-aware Prior

(a) Input image I　(b) Shading by CRefNet [29]　(c) Prior-guided shading S'　(d) Recovered shading S

Quantitative Comparison and Applications

Method	WHDR (IIW)	Running time (512×384)
Retinex-color [3]	26.89%	250.30s
Zhao *et al.* [2]	23.20%	7.11s
Bell *et al.* [1]	20.64%	209.00s
Fu *et al.* [11]	19.20%	9.60s
Li *et al.* [24]	15.40%	-
NIID-Net [32]	16.60%	-
CRefNet [29]	14.50%	-
Ours	**13.62%**	**2.27s**



Input　Low-light/Recoloring

Please see https://github.com/xiaodongwo/Efficient_IID